

Trajectory Prediction and ROS-based Control for a Heterogeneous Robot Arm via Local Fine-Tuning of the Pi-Zero VLA Model

사전학습된 Vision-Language-Action 모델을 사전학습에 포함되지 않은 6-DoF 로봇팔에 전이하고, ROS 기반 실시간 제어로 실제 pick-and-place를 검증한다

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VLA

Pi-Zero

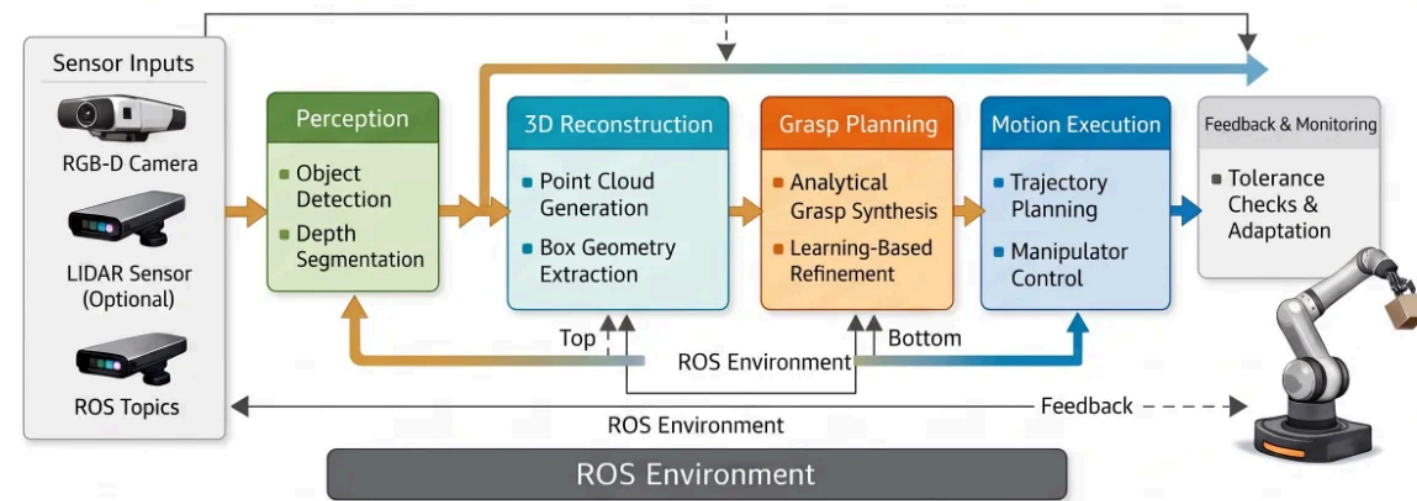
PEFT · LoRA

Robot manipulation

ROS

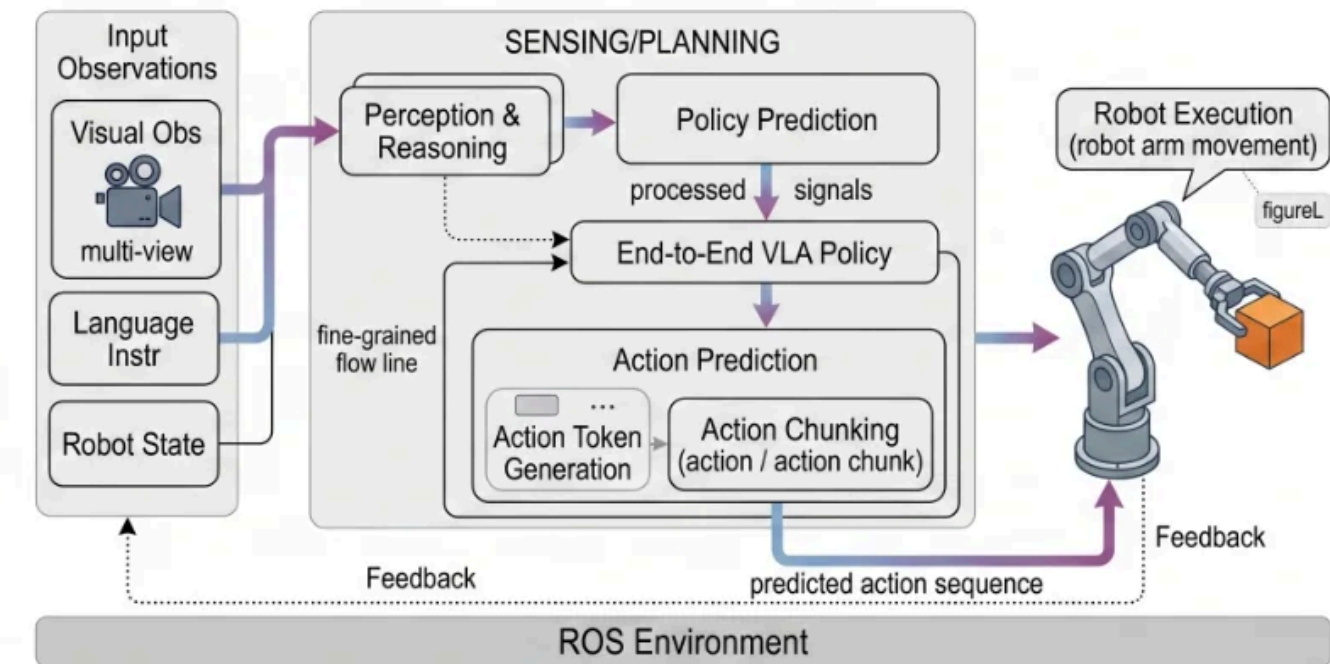
From Modular Perception-Planning-Control to End-to-End VLA

기존 모듈형 로봇 조작 파이프라인



인식·계획·제어 모듈을 개별 설계하여 연산 지연과 오차 누적이 발생하기 쉽다.

VLA End-to-End 제어

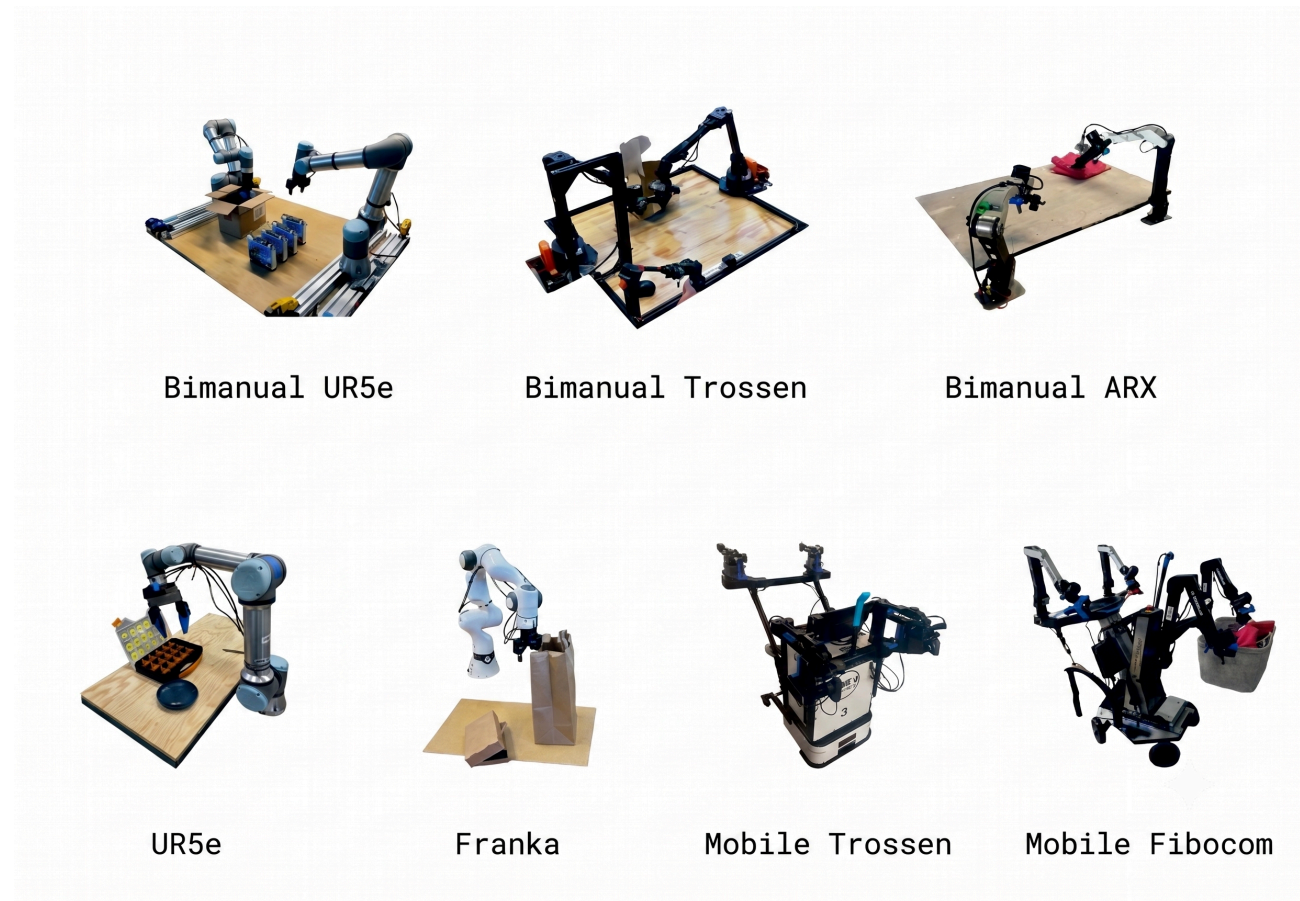


이미지와 언어 명령으로부터 로봇 행동을 직접 예측하여 유연하고 지능적인 작업 수행이 가능하다.

Embodiment Gap for a Robot Absent from Pretraining

6-DoF Dobot Magician E6는 Pi-Zero의 사전학습 구성에 포함되지 않아, 기하학적 차이가 성능 저하와 제어 불안정으로 이어질 수 있다.

Pi-Zero 사전학습 로봇 · 포함됨



≠
Embodiment
gap

본 연구 대상 · Dobot Magician E6 · 미포함



Full fine-tuning 한계

대규모 GPU 메모리 · 학습 시간

Edge 환경의 높은 메모리·연산 부담

Catastrophic forgetting

→ 선택적 PEFT 필요

Related Work — VLA and Efficient Transfer

RT-1 · RT-2	대규모 로봇 데이터와 VLM 의미 지식으로 다양한 객체·언어 명령에 대한 일반화	[1,2]
Open-X · Octo · OpenVLA	이기종(cross-robot) 정책 학습과 새로운 센서·행동 공간에 대한 fine-tuning	[3-5]
Pi-Zero (π_0)	사전학습된 VLM 위에 flow matching 기반 Action Expert를 결합한 VLA 모델	[6]
LoRA · Action chunking	Parameter-efficient fine-tuning과 action chunking이 효율적 로봇 적응 방법으로 제시됨	[7-10]

System Overview

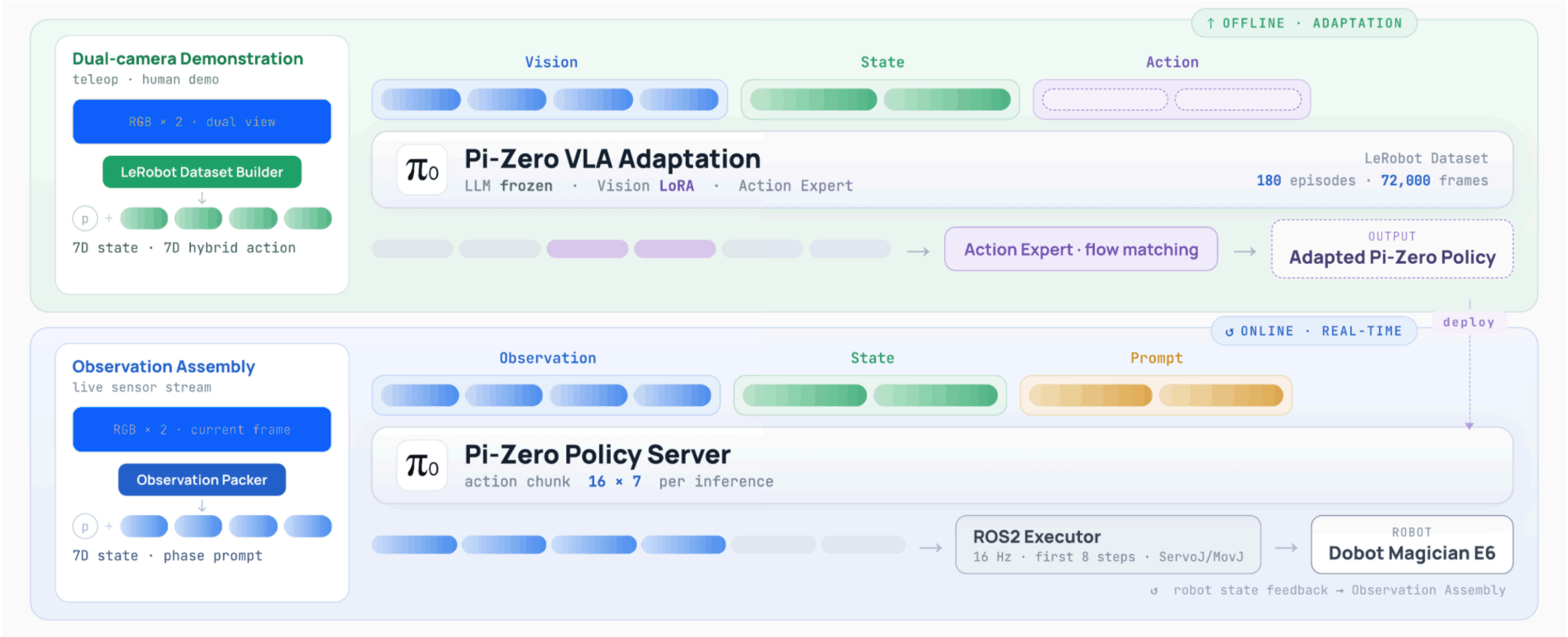


Fig. 2 Overall architecture of local Pi-Zero VLA adaptation and ROS2-based real-time execution

Local Adaptation of the Pi-Zero VLA

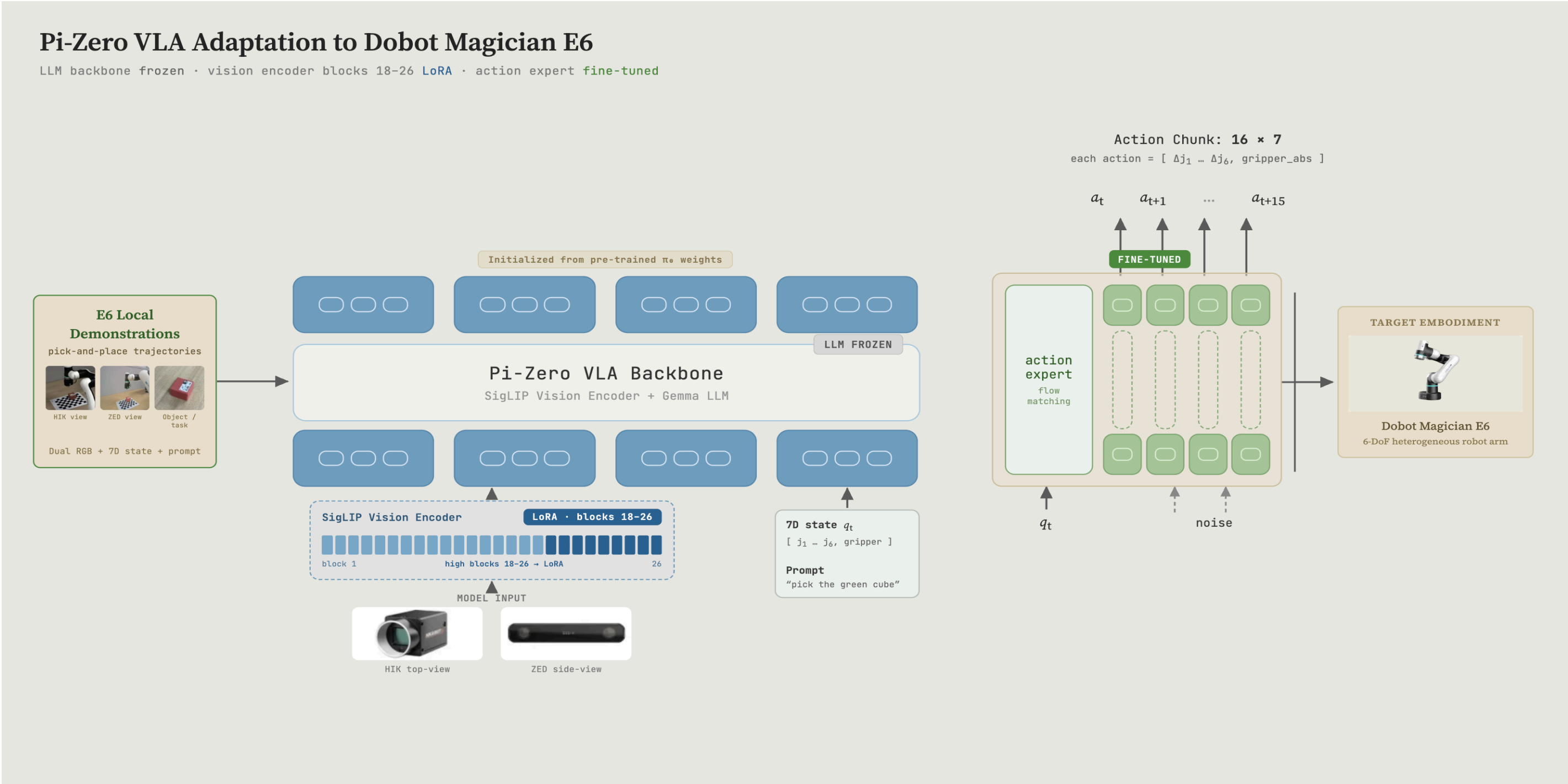


Fig. Local LoRA fine-tuning and Action Expert adaptation of Pi-Zero

Dual-Camera Dataset Construction



Fig. 1 Data collection setup

Episodes	180
Frames / episode	400
Total frames	72,000
Camera views	HIK top / ZED side
Image / format	224×224 RGB × 2 views / LeRobot

State = [j1, j2, j3, j4, j5, j6, gripper]

Action = [Δj1, Δj2, Δj3, Δj4, Δj5, Δj6, gripper_abs]

7D state · 7D hybrid action (joint delta + absolute gripper)

180 episodes × 400 frames = 72,000 frames.

ROS2-based Real-time Execution Pipeline

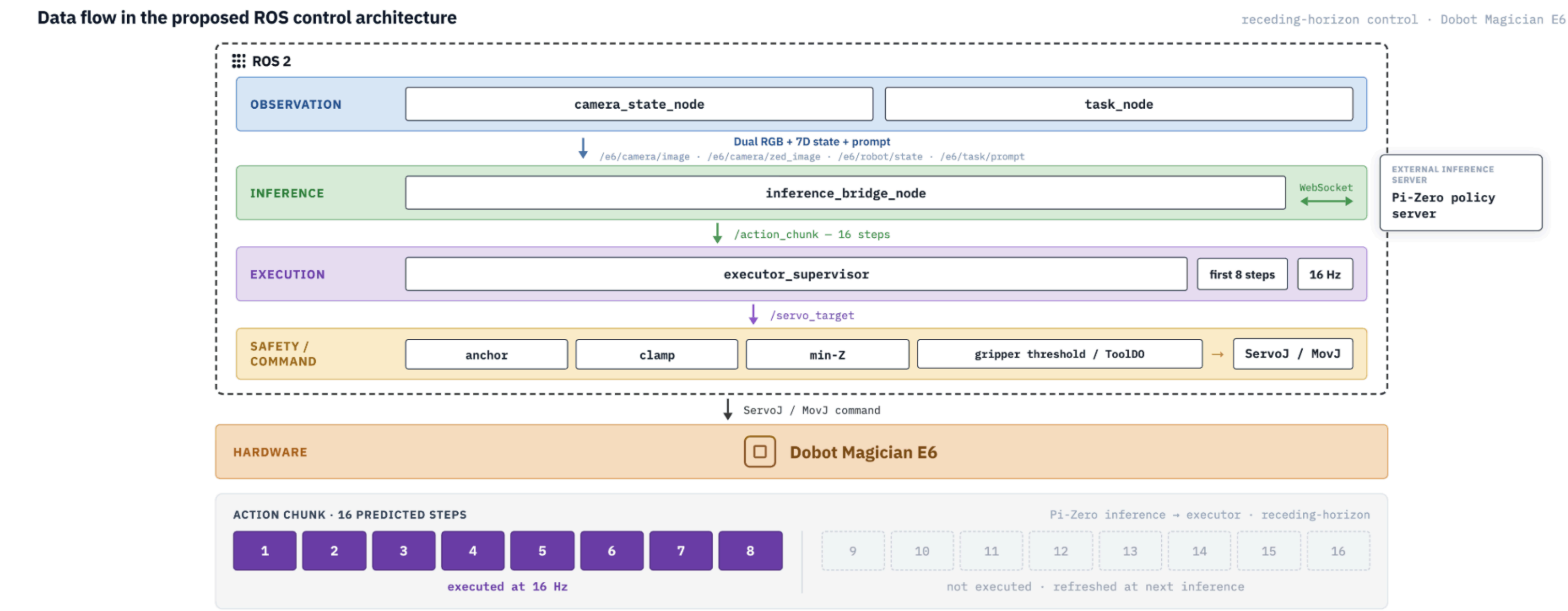


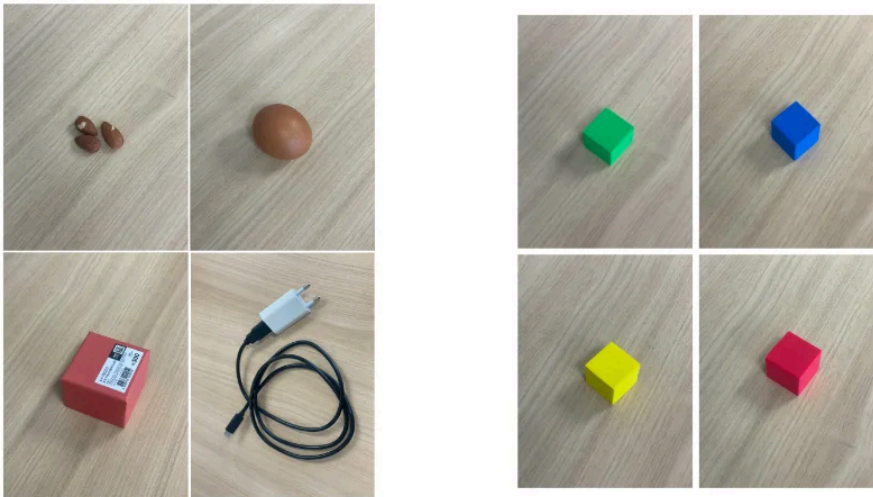
Fig. 3 ROS2-based real-time execution pipeline converting Pi-Zero action chunks into Dobot Magician E6 control commands.

Experimental Conditions and Evaluation Criteria

Test	Condition	Trials	Evaluation
Object recognition	single target object	20	target object picked
Multi-object recognition	similar / mixed objects	per condition	commanded object selected
Color recognition	same shape, diff. color	20 each	commanded color picked
Pick-and-place	random placement	20	pick-and-place completed
Control comparison	ServoJ vs MovJ	repeated	completion time / RTT

Success was determined by whether the commanded object or color was correctly picked and placed in the real robot environment.

Task Success Results

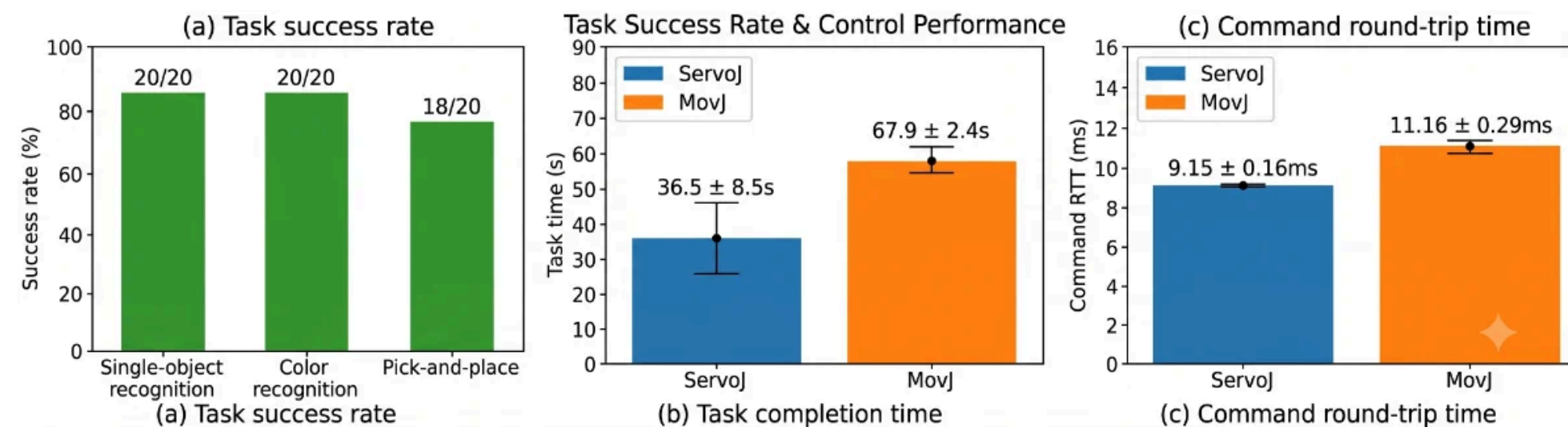


Test	Result	Interpretation
Single-object recognition	20 / 20	object grounding retained
Color recognition	20 / 20	language-conditioned selection
Pick-and-place	18 / 20	real-world manipulation

사전학습에 포함되지 않은 6-DoF 로봇팔로의 전이 가능성을 확인했다.
다만 아몬드·계란처럼 형태가 유사한 소형 객체 간에는 혼동이 관찰되어, 세밀한 객체 구분에는 한계가 있었다.

Fig. 3 Test objects and color targets

Control Efficiency: ServoJ vs MovJ



Task completion time
36.5 s vs 67.9 s

Round-trip delay
9.15 ms vs 11.16 ms

ServoJ는 MovJ 대비 작업 완료 시간과 통신 지연을 모두 줄인다.

→ 실시간 16 Hz action-chunk 실행에는 ServoJ 방식이 더 적합하다.

Fig. 4 Task completion time and round-trip delay — ServoJ vs MovJ

Validating Heterogeneous Robot-Arm Transfer via Lightweight Local Fine-Tuning

01

Transferred the Pi-Zero VLA model to a 6-DoF Magician E6 arm.

02

Constructed a dual-camera dataset with 7D hybrid actions.

03

Validated ROS-based 16 Hz real-world pick-and-place control.

제한된 로컬 데이터와 경량 미세조정만으로 사전학습에 포함되지 않은 로봇팔에 VLA 전이가 가능함을 확인
— 임베디드 환경에서의 실시간 Physical AI 구현 가능성에 대한 초기 검증으로서 의미를 가진다.

References

[\[1\]](#) A. Brohan et al., "RT-1: Robotics Transformer for Real-World Control at Scale," arXiv:2212.06817, 2022.

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[\[3\]](#) Open X-Embodiment Collaboration et al., "Open X-Embodiment: Robotic Learning Datasets and RT-X Models," arXiv:2310.08864, 2023.

[\[4\]](#) Octo Model Team et al., "Octo: An Open-Source Generalist Robot Policy," arXiv:2405.12213, 2024.

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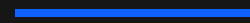
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Q&A



Thank you for your attention